KIRAN POUDEL

+977 9864483144 | kiranpoudel160@gmail.com | Kirtipur-04, Kathmandu

CAREER SUMMARY

Computer Engineering graduate with professional experience in ASP.NET development and strong research interests in machine learning, behavioral analytics, and intelligent systems. Experienced in developing ML-driven applications and conducting data-centric research involving clustering, NLP, and statistical modeling. Research interests include Machine Learning, Natural Language Processing, Statistical Learning, Explainable AI, and Predictive Analytics.

EDUCATION

- Computer Engineering | Everest Engineering College, Sanepa, Lalitpur — CGPA: 3.34 | 2025 AD |
- Higher Education(+2) | Gyanodaya Secondary School, Bafal, Kathmandu — CGPA: 3.61 | 2020 AD |
- Secondary Education(S.E.E) | Gyanodaya Secondary School, Bafal, Kathmandu — CGPA: 3.5 | 2018 AD |

TECHNICAL SKILLS

- **Programming Languages**: Python, C, C++, JavaScript (ES6+), TypeScript, C#
- **Frontend Development**: HTML5, CSS3, React.js, Blazor
- **Backend & Frameworks**: ASP.NET (.NET), Node.js, Express.js, Laravel (PHP)
- **Databases**: MS SQL Server (Stored Procedures), MySQL, MongoDB
- **Concepts & Tools**: REST APIs, Git, JWT Authentication, CORS, Custom Middleware, Socket.IO

EXPERIENCE

- **URANUS TECH PVT. LTD. (ASP.NET Developer) | *Jan 2026 – Present***
  - Developing and maintaining web applications using ASP.NET, focusing on efficient backend logic and structured code practices.
  - Working with SQL Server for database operations, including query optimization and data management.
  - Implementing business logic and features based on real-world client requirements within an enterprise environment.
- **Broadway Infosys (MERN Stack Training – 3 Months(aug-nov 2025))**
  - Completed hands-on MERN stack training (MongoDB, Express.js, React.js, Node.js) with focus on building scalable full-stack applications.
  - Designed and developed a gig-based web platform featuring secure user authentication, dynamic service listings, and interactive client provider workflows.

ACADEMIC PROJECTS

- **IELTS Essay Evaluation Model (AI/ML Project**) |*Python, MERN, LLM, Hugging Face, Fine tuning*
  - Developed an NLP-based IELTS essay evaluation system using LSTM and fine-tuned Mistral-7B models on structured writing datasets to generate automated band-score predictions and feedback.
  - Performed text preprocessing such as tokenization, stop-word removal, and normalization. Designed the system to generate structured feedback and band scores based on IELTS criteria, applying prompt-engineering techniques to enhance accuracy.

- **Fear of Missing Out (FOMO) in Engineering Students (Research Project**) | Python, Machine Learning, SEM, Network Analysis
  - Conducted a large-scale survey-based study to analyze psychological factors influencing FOMO among engineering students.
  - Applied clustering techniques, Random Forest classification, and Statistical Modeling including Structural Equation Modeling (SEM) and Mixed Graphical Models (MGM) to identify high-risk student profiles.
  - Built a Psychological Network Analysis (PNA) model to examine relationships between constructs such as social comparison, peer pressure, academic burnout, loneliness, and life satisfaction, providing data-driven insights into student behavior.

- **Email Spam Detection (ML Project**) |*Python, MERN, Random Forest, Hugging Face*
  - Developed a spam-detection system using a Hugging Face dataset and trained a Random Forest classifier to classify emails as spam or not spam.
  - Built a MERN-based messaging system where multiple users can send and receive messages, and each message is processed by the ML model for spam detection.

OTHER RESEARCH PROJECTS

- **Data-Driven Waste Management System**
  - Developed a demand-driven smart waste management platform integrating real-time waste reporting, dynamic vehicle allocation, and route optimization to improve collection efficiency.
  - Optimized waste collection routing using clustering and TSP heuristics, reducing estimated travel distance and operational redundancy in simulated municipal collection scenarios.
  - Integrated household waste segregation using color-coded bins for Organic Waste and Non-Organic Waste, along with gamification features to improve citizen participation and sustainable waste management.
- **FakeCure Detector**
  - Developed an NLP-based system that analyzes user-described symptoms to predict potential health conditions and suggest relevant medical information using trained classification models.
  - Integrated explainable AI techniques to provide transparent reasoning behind predictions, helping users understand how symptom patterns relate to possible conditions.

- **Climate Change Topic Modeling and Organizational Research Analysis**
  - Developed a topic modeling framework to analyze global climate change research papers from different organizations and institutions.
  - Identified year-wise topic trends to determine when specific climate issues received the highest research attention across regions and sectors.
  - Evaluated whether organizations' published research aligned with their stated climate goals, priorities, and real-world environmental initiatives.

(*Research manuscripts based on these projects are currently under preparation for future conference and journal submission*.)